Quiz 3 (20mins)

Q1. Answer the following in 1-2 lines. (3)

(a) What is the bandgap, electron affinity, intrinsic carrier concentration of Si at 300K?

(b) 300K would lead to thermal energy (kT) of ~25meV whereas to generate an electron-hole pair, at least energy equal to the bandgap (>>25meV) would be required. How come we get intrinsic carrier concentration?

Q2. Plot the extrinsic carrier concentration for Si with Temperature. Describe the dependence and describe dominant source of carriers (3)

Q3. If the Silicon is doped with $N_d=1\times10^{16}$ donors/cm³

a. What are the equilibrium electron and hole concentrations? (2)

b. Draw band diagram and mark all the values in band diagram (2)

$$n = n_i e^{\frac{E_F-E_i}{kT}}$$

Solution 1: (a) bandgap of Si: ~1.12eV

Electron affinity of Si:~4.05eV

Intrinsic carrier concentration at 300k: ni=~(1 to 1.5)$\times10^{10}$ cm^{-3}

(b) Although kT ≈ 25 meV at 300 K is much smaller than the bandgap, electrons follow a Fermi–Dirac (Boltzmann) energy probability distribution function (due to the wave nature of electron), so a small fraction of electrons in the high-energy tail acquire sufficient energy to cross the bandgap, resulting in a finite intrinsic carrier concentration. Also ,at 300K, 25meV is the average energy available, there may be electrons in the valence band that possess energies higher than this 25meV and crosses from Valence band to conduction band.

Solution 2:

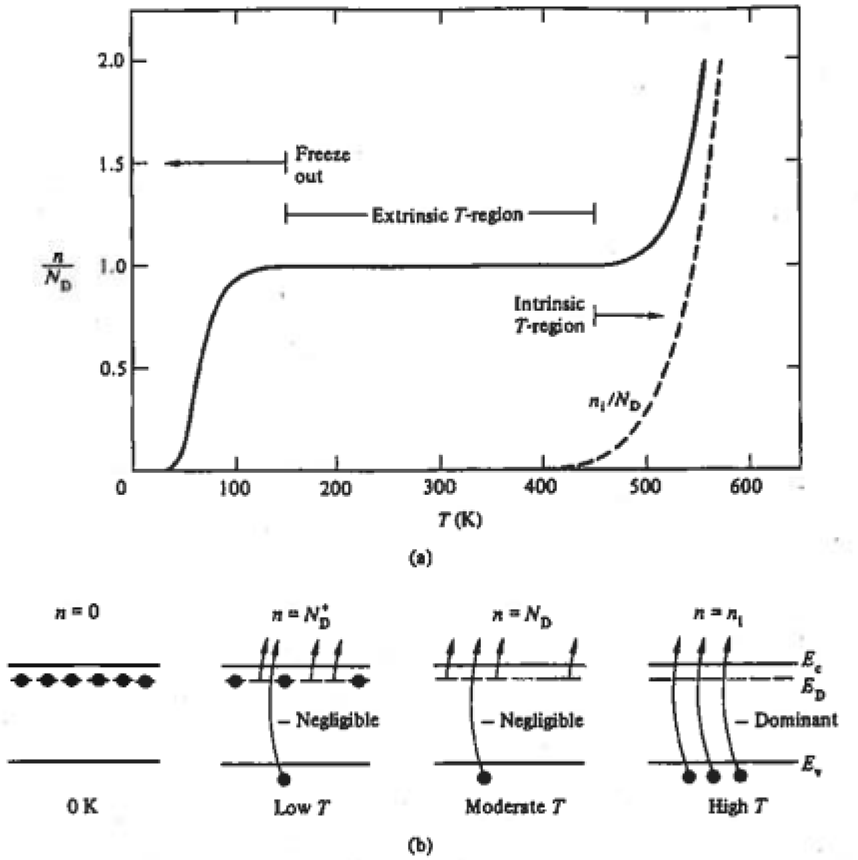


Figure 2.22 (a) Typical temperature dependence of the majority-carrier concentration in a doped semiconductor. The plot was constructed assuming a phosphorus-doped $N_D = 10^{15}/\text{cm}^3$ Si sample. n_i/N_D versus T (dashed line) has been included for comparison purposes. (b) Qualitative explanation of the concentration-versus-temperature dependence displayed in part (a).

Explanation of Dependence on Temperature

1. Freeze-out Temperature Region (Low Temperature)

At very low temperatures ($T \to 0K$), the thermal energy available in the semiconductor is insufficient to ionize donor atoms. Hence, electrons remain weakly bound to donor sites and the free electron concentration is much less than the donor concentration. $n \ll ND$

Dominant source of carriers: Ionized donors (very few).

2. Extrinsic Temperature Region (Moderate Temperature)

As the temperature increases, almost all donor atoms become ionized. In this region, the electron concentration becomes nearly constant and approximately equal to the donor concentration. This region represents the normal operating range of most solid-state devices. $n \approx ND$

Dominant source of carriers: Dopant atoms

3. Intrinsic Temperature Region (High Temperature)

At sufficiently high temperatures, thermal generation of electron–hole pairs becomes significant. The intrinsic carrier concentration increases rapidly and eventually exceeds the dopant-supplied carriers. The semiconductor then behaves like an intrinsic material.

$n \to n_i$ and $n_i > N_D$

Dominant source of carriers: Thermally generated electron–hole pairs

Solution 3:

Q(3)

(a) Si doped with $N_d = 10^{16}$ donors $/cm^3$
n-type

$n_i \approx 10^{10}/cm^3$

If donor conc or acceptor conc is much much higher than intrinsic conc then the majority conc = doping Concentration

$\cong \boxed{10^{16}/cm^3 = e^- \text{ Conc}^n}$

Holes $\simeq p = \dfrac{n_i^2}{n} = \dfrac{10^{20}}{10^{16}} = 10^4/cm^3$

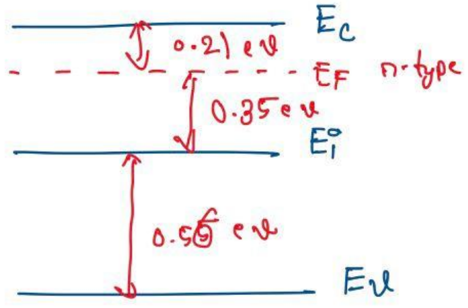
(b) Band diagram :

$n = n_i e^{(E_F - E_i)/KT}$

$10^{16} = 10^{10} e^{(E_F - E_i)/KT}$

$10^6 = e^{(E_F - E_i)/KT}$

$13.81 = \dfrac{E_F - E_i}{KT}$



$KT \approx 25.85\,meV$

$E_F - E_i = 357\,meV$

$\approx 0.35\,eV$